Houssam Zenati

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Research Interests

I develop machine learning algorithms and statistical theory for learning from partial feedback, understanding intervention effects, and designing adaptive experiments. My research progresses from learning which interventions to choose, to establishing what they change, to deciding what to try next—when good decisions and informative experiments need not favor the same assignments.

Professional Experience

Gatsby Computational Neuroscience Unit, University College London, United Kingdom
Research Fellow 2025–present
I work with Arthur Gretton.

MIND team, Inria, France
Postdoctoral Researcher 2023–2024
Research on causal inference and mediation analysis.

Criteo AI Lab, France
PhD Researcher 2020–2023
Research Engineer 2019–2020
PhD jointly with Inria Thoth and Criteo AI Lab on offline policy learning and sequential learning.

Institute for Infocomm Research (A*STAR), Singapore 2017–2018
Research Engineer
Research Intern
Worked on deep generative models and anomaly detection.

Education

Université Grenoble Alpes / INRIA Thoth, France 2023
Ph.D. in applied mathematics and machine learning.
Thesis: *Efficient Methods in Counterfactual Policy Learning and Sequential Decision Making.*

ENS Paris-Saclay / MVA, France 2019
M.Sc. in mathematics, vision, and learning.

CentraleSupélec, France 2015–2019
M.Eng. in applied mathematics and computer science.

Selected Publications

H. Zenati, E. Diemert, M. Martin, J. Mairal, P. Gaillard (2023). *Sequential Counterfactual Risk Minimization.* ICML.
ICML 2023 proceedings

H. Zenati, B. Bozkurt, A. Gretton (2025). *Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings.* NeurIPS.
NeurIPS 2025 proceedings

H. Zenati, A. Bietti, E. Diemert, J. Mairal, M. Martin, P. Gaillard (2022). *Efficient Kernelized UCB for Contextual Bandits.* AISTATS.
AISTATS 2022 proceedings

Preprints

∗ marks authors who contributed equally to the paper.

B. Bozkurt, H. Dance, Z. Chen, A. Galashov, A. Gretton, **H. Zenati** (2026). *Sampling Interventional Distributions with Orthogonal MMD Flow*. Preprint.

R. Leman, B. Bozkurt, S. Tanji, A. Gretton, **H. Zenati** (2026). *Doubly Robust Deep Kernel Tests for Structured Outcomes*. Preprint.

F. El Khoury*, **H. Zenati***, N. Kallus, M. Arbel, A. Bibaut (2026). *Parametric Rates for Nonparametric Conditional-Moment Bilevel Optimization*. Preprint.

B. Bozkurt, A. Galashov, D. Meunier, Z. Shen, A. Gretton, **H. Zenati** (2026). *Doubly Robust Proxy Causal Learning with Neural Mean Embeddings*. Preprint.

Z. Shen, N. Kallus, D. Meunier, **H. Zenati**, A. Gretton, A. Bibaut (2026). *Nonparametric Instrumental Variable Analysis Without Structural Equations: Debiased Inference on Functionals of Inverse Problems with No Solutions*. Preprint.

A. Bibaut, **H. Zenati**, T. Rahier, N. Kallus (2026). *Functional Natural Policy Gradients*. Preprint.

Z. Shen*, **H. Zenati***, N. Kallus, A. Gretton, K. Khamaru, A. Bibaut (2026). *Efficient Inference after Directionally Stable Adaptive Experiments*. Preprint.

Other Publications

† marks alphabetical author order.

H. Zenati, B. Bozkurt, A. Gretton (2026). *Kernel Treatment Effects with Adaptively Collected Data*. AISTATS.

S. Girard, A. Bibaut, J.-J. Vie, A. Gretton, N. Kallus, **H. Zenati** (2026). *Fast Best-in-Class Regret for Contextual Bandits*. UAI.

H. Zenati, A. Gretton (2026). *Semiparametric Efficient Tests for Interpretable Distributional Treatment Effects*. NeurIPS.

H. Zenati, J. Abecassis, J. Josse, B. Thirion (2025). *Double Debiased Machine Learning for Mediation Analysis with Continuous Treatments*. AISTATS.

B. Bozkurt, **H. Zenati**, D. Meunier, L. Xu, A. Gretton (2025). *Density Ratio-Free Doubly Robust Proxy Causal Learning*. NeurIPS.

J. Abecassis, **H. Zenati**, S. Boumaiza, J. Josse, B. Thirion (2025). *Causal Mediation Analysis with One or Multiple Mediators: A Comparative Study*. Psychological Methods.

H. Zenati, A. Bietti, M. Martin, E. Diemert, P. Gaillard, J. Mairal (2025). *Counterfactual Learning of Stochastic Policies with Continuous Actions*. Transactions on Machine Learning Research.

J. Zhou, P. Gaillard, T. Rahier, **H. Zenati**, J. Arbel (2024). *Towards Efficient and Optimal Covariance-Adaptive Algorithms for Combinatorial Semi-Bandits*. NeurIPS.

M. Martin, P. Mertikopoulos, T. Rahier, **H. Zenati** (2022). *Nested Bandits*†. ICML.

H. Zenati, A. Bietti, M. Martin, E. Diemert, J. Mairal (2020). *Optimization Approaches for Counterfactual Risk Minimization with Continuous Actions*. ICLR Workshop on Causal Learning for Decision Making.

P. Mertikopoulos, B. Lecouat, **H. Zenati**, C.-S. Foo, V. Chandrasekhar, G. Piliouras (2019). *Optimistic Mirror Descent in Saddle-Point Problems: Going the Extra (Gradient) Mile*. ICLR.

K. Ouardini, H. Yang, B. Unnikrishnan, M. Romain, C. Garcin, **H. Zenati**, J. P. Campbell, M. F. Chiang, J. Kalpathy-Cramer, V. Chandrasekhar, et al. (2019). *Towards Practical Unsupervised Anomaly Detection on Retinal Images*. MICCAI Workshop on Domain Adaptation and Representation Transfer.

H. Zenati, M. Romain, C.-S. Foo, B. Lecouat, V. Chandrasekhar (2018). *Adversarially Learned Anomaly Detection*. ICDM.

H. Zenati, C.-S. Foo, B. Lecouat, G. Manek, V. R. Chandrasekhar (2018). *Efficient GAN-Based Anomaly Detection*. ICLR Workshop.

B. Lecouat, C.-S. Foo, **H. Zenati**, V. Chandrasekhar (2018). *Semi-Supervised Learning with GANs: Revisiting Manifold Regularization*. ICLR Workshop.

B. Lecouat, C.-S. Foo, **H. Zenati**, V. Chandrasekhar (2018). *Manifold Regularization with GANs for Semi-Supervised Learning*. ICLR Workshop.

B. Lecouat, K. Chang, C.-S. Foo, B. Unnikrishnan, J. M. Brown, **H. Zenati**, A. Beers, V. Chandrasekhar, J. Kalpathy-Cramer, P. Krishnaswamy (2018). *Semi-Supervised Deep Learning for Abnormality Classification in Retinal Images*. NeurIPS Workshop on Machine Learning for Health.

Teaching

Introduction to Causal Inference with Hidden Confounders: Instruments and Proxies, UM6P — 12 November 2025
Guest lecture. A 90-minute lecture followed by a separate 90-minute practical, progressing from potential outcomes and adjustment with observed covariates to instruments, proxy variables, and bridge functions for hidden confounding. Slides.

Causal Inference, New York University Paris — 2024–2025
Co-instructor, with Judith Abecassis. This course introduces the main tools of causal inference, from randomized experiments to observational and quasi-experimental methods, with a particular emphasis on observational inference.

Advanced Machine Learning, CentraleSupélec — 2024–2025
Teaching assistant, alongside Emilie Chouzenoux. This course covers modern machine learning methods at the interface of statistics and optimization, with an emphasis on connecting theory, assumptions, and algorithmic practice.

Data Visualisation, Université Paris Dauphine – PSL — 2021–2022
Instructor for Master's students in the IASD program, presenting data visualisation as a statistical and algorithmic tool for exploring, summarizing, and communicating complex data.

When teaching, I like to emphasize conceptual clarity, mathematical rigor, and active engagement, with the goal of helping students understand both the structure of a method and the assumptions that make it work.

Student Supervision

I have been fortunate to co-supervise and mentor students on research projects:

Julien Zhou, Master's student, mentoring, with Pierre Gaillard — 2023
Towards Efficient and Optimal Covariance-Adaptive Algorithms for Combinatorial Semi-Bandits. NeurIPS 2024.

Samuel Girard, PhD student, co-supervision, with Aurelien Bibaut — 2025–2026
Fast Best-in-Class Regret for Contextual Bandits. UAI 2026.

Robin Leman, Master's student, supervision, with Arthur Gretton — 2026
Doubly Robust Deep Kernel Tests for Structured Outcomes. Preprint, 2026.

Bariscan Bozkurt, PhD student, co-supervision, with Arthur Gretton — 2026
Doubly Robust Proxy Causal Learning with Neural Mean Embeddings. Preprint, 2026.
Sampling Interventional Distributions with Orthogonal MMD Flow. Preprint, 2026.

Sofiane Tanji, PhD student, co-supervision, with Yassine Laguel — 2026
Work in progress.

Service

Workshop organisation
Lead Organiser, Advances in Adaptive Experimentation (AAE) Workshop, Gatsby Computational Neuroscience Unit, UCL, 18–19 June 2026, with Bariscan Bozkurt, I-Chun Lin and Arthur Gretton. The two-day programme included seven invited speakers, posters and an open-problems session.

Seminar organisation
Organiser of the external machine learning seminars at Gatsby Unit, UCL, September 2025–August 2026.

Reading groups
Co-organized, with Dimitri Meunier, a reading group at Gatsby, UCL on Aad van der Vaart's *Asymptotic Statistics*, 2026.
Organized, with Fateme Ghayem, a reading group at Inria Paris on Martin Wainwright's *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, 2023–2024.

Peer review
NeurIPS, ICML, ICLR, AISTATS and TMLR.

Funding and Awards

Netflix workshop grant 2026
Secured US$5,000 from Netflix to support the Advances in Adaptive Experimentation (AAE) Workshop at Gatsby, UCL, 18–19 June 2026.

IEEE ICDM Student Travel Award 2018
Support for conference participation.

Singapore International Pre-Graduate Award (SIPGA) 2017
Support for a research attachment at A*STAR, Singapore.

Selected Talks

Interventional Mean Embeddings for Distributional Causal Inference, Gatsby Unit, UCL, London, United Kingdom — Oct. 2026

Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings, Symposium on Mathematical Foundations of Trustworthy Learning, Monte Verità, Ascona, Switzerland — Oct. 2025

Understanding Influence Functions, Gatsby Machine Learning Journal Club, Gatsby Unit, UCL, London, United Kingdom — Jan. 2025

Sequential Counterfactual Risk Minimization, ELLIS UnConference, HEC Paris, Jouy-en-Josas, France — Jul. 2023

Nested Bandits, ICML, Baltimore, United States — Jul. 2022

Efficient Kernel UCB for Contextual Bandits, Thoth seminar, Inria, Grenoble, France — Jan. 2022

Counterfactual Learning of Stochastic Policies with Continuous Actions, MIPT–UGA AI Workshop, online — Jul. 2021

Optimization Approaches for Counterfactual Risk Minimization with Continuous Actions, ICLR Workshop on Causal Learning for Decision Making, online — Apr. 2020

Adversarially Learned Anomaly Detection, ICDM, Singapore — Nov. 2018

Software

Open-source library

med_bench — Psychological Methods 2025
Python package for causal mediation analysis with common estimators and flexible nuisance models.

Research software

adaptive-KTE — AISTATS 2026
Kernel treatment effects with adaptively collected data.

Interpretable distributional treatment effects — NeurIPS 2026
Semiparametrically efficient local testing with learned outcome locations.

Continuous-treatment mediation — AISTATS 2025
Double debiased machine learning for mediation analysis with continuous treatments.

Counterfactual policy mean embeddings — NeurIPS 2025
Doubly robust estimation of counterfactual policy mean embeddings.

Sequential counterfactual risk minimization — ICML 2023
Sequential counterfactual risk minimization for policy learning.

Efficient Kernel UCB — AISTATS 2022
Efficient kernelized UCB for contextual bandits.

Nested bandits — ICML 2022
Nested exponential weights for nested bandits.

Continuous-action counterfactual risk minimization — ICLR Workshop 2020; TMLR 2025
Counterfactual learning and optimization of stochastic policies with continuous actions.

Adversarially learned anomaly detection — ICDM 2018
Adversarially learned models for anomaly detection.

Efficient GAN anomaly detection — ICLR Workshop 2018
Efficient GAN-based anomaly detection.